

ASTRODYNAMICS HW#2 SOLUTIONS

For the following problems, use $\mu=398,600.4418\text{km}^3/\text{sec}^2$ and $R_E=6,378\text{km}$ if necessary.

1. Calculate the eccentric anomaly and true anomaly for $M = 215.4^\circ$ and $e = 0.3$.

$$E_{n+1} = E_n + \frac{M - E_n + e \sin(E_n)}{1 - e \cos(E_n)}$$

$$\text{Begin by letting } E_1 = M - e = 215.4^\circ \times \frac{\pi}{180^\circ} - 0.3 = 3.4577 \text{ [rad]}$$

E_n	
E_1	3.4577
E_2	3.6186
E_3	3.5370
E_4	3.6197
E_5	3.6197

E_n converge on 3.6197

2. A rogue asteroid has just been discovered. Your job is to determine how much time we have to intercept and destroy the object, if $a = -2,797.425\,069\text{km}$, $e = 2.8$, $i = 23^\circ$, $\nu = 249.27^\circ$, and $r = 2,150,138.804\,411\text{km}$.

$$r_0 = 2,150,138.804\,411 \text{ km}$$

$$\nu_0 = 249.27^\circ = 4.350\,582 \text{ [rad]}$$

$$\sinh H_0 = \frac{\sin(\nu_0) \sqrt{e^2 - 1}}{1 + e \cos(\nu_0)} = -274.860\,162$$

$$H_0 = -6.309\,413$$

If we use general method of TOF, then

$$r = 6378.136\,300 \text{ km}$$

$$\nu = \arccos\left(\frac{1}{e}\left(\frac{a(1 - e^2)}{r} - 1\right)\right) = \arccos\left(\frac{1}{e}\left(\frac{(-2,797.425\,069\text{km})(1 - 2.8^2)}{6378.136\,300\text{km}} - 1\right)\right) = 315.584\,597^\circ$$

$$\sinh H = \frac{\sin(\nu) \sqrt{e^2 - 1}}{1 + e \cos(\nu)} = -0.610\,120$$

$$H = -0.577\,484$$

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$$t - T = \frac{1}{n}(e \sinh H - H), \quad n = \sqrt{\frac{\mu}{|a|^3}}$$

$$\begin{aligned} t - t_0 &= \frac{1}{n}[(e \sinh H - H) - (e \sinh H_0 - H_0)] \\ &= 178,615.706\,558s \end{aligned}$$

3. You have received two position vectors for a satellite. How much time has elapsed between these two positions? ($p = 6,681.571km$)

$$\vec{r}_0 = -2,574.9533\hat{i} + 4,267.0671\hat{j} + 4,431.5026\hat{k}$$

$$\vec{r} = 2,700.6738\hat{i} - 4,303.5378\hat{j} - 4,358.2499\hat{k}$$

$$r_0 = |\vec{r}_0| = 6693.909079km$$

$$r = |\vec{r}| = 6669.067508km$$

$$\cos(\Delta v) = \frac{\vec{r}_0 \cdot \vec{r}}{r_0 r} = -0.999754888$$

$$\begin{aligned} k &= r_0 r (1 - \cos(\Delta v)) = 6669.067588km \cdot 6693.909079km \cdot (1 - (-0.999754888)) \\ &= 8.927333207 \times 10^7 km^2 \end{aligned}$$

$$l = r_0 + r = 1.336297658 \times 10^4 km$$

$$\begin{aligned} m &= r_0 r (1 + \cos(\Delta v)) = 6669.067588km \cdot 6693.909079km \cdot (1 + (-0.999754888)) \\ &= 1.094230523 \times 10^4 km^2 \end{aligned}$$

$$a = \frac{m k p}{(2m - l^2)p^2 + 2klp - k^2} = 6681.607753295km$$

$$f = 1 - \frac{r}{p}(1 - \cos(\Delta v)) = 1 - \frac{6693.909079km}{6681.571km}(1 - (-0.999754888)) = -1.003447602$$

$$g = \frac{r_0 r \sin(\Delta v)}{\sqrt{\mu p}} = \frac{6669.067588km \cdot 6693.909079km \cdot 0.022139628}{\sqrt{3.986004418km^3s^{-2} \cdot 6681.754km}} = 19.151688789s$$

for elliptical orbit ($a > 0$)

$$\tan^2\left(\frac{\Delta v}{2}\right) = \frac{1 - \cos(\Delta v)}{1 + \cos(\Delta v)} = \frac{1 - (-0.999754888)}{1 + (-0.999754888)} = 8158.535233$$

$$\tan\left(\frac{\Delta v}{2}\right) = -90.32461034$$

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$$\begin{aligned}\dot{f} &= \sqrt{\frac{\mu}{p}} \tan\left(\frac{\Delta v}{2}\right) \left\{ \frac{1 - \cos(\Delta v)}{p} - \frac{1}{r_0} - \frac{1}{r} \right\} \\ &= \sqrt{\frac{3.986004418 \times 10^5 \text{ km}^3/\text{s}^2}{6681.754 \text{ km}}} (-90.32461034) \left\{ \frac{1 - (-0.999754888)}{6681.754 \text{ km}} \right. \\ &\quad \left. - \frac{1}{6669.067508 \text{ km}} - \frac{1}{6693.909079 \text{ km}} \right\} = -2.889964677 \times 10^{-5} \text{ s}^{-1}\end{aligned}$$

$$\begin{aligned}\cos(\Delta E) &= 1 - \frac{r_0}{a}(1 - f) = 1 - \frac{6669.067508 \text{ km}}{6681.754 \text{ km}} (1 - (-1.003447602)) = -0.999687548 \\ \sin(\Delta E) &= -\frac{r_0 r \dot{f}}{\sqrt{\mu a}} = -\frac{6669.067508 \text{ km} \cdot 6693.909079 \text{ km} \cdot -2.889964677 \times 10^{-5} \text{ s}^{-1}}{\sqrt{3.986004418 \text{ km}^3/\text{s}^2} \cdot 6681.754 \text{ km}} = 0.024999313 \\ \Delta E &= 3.116590735 \text{ rad}\end{aligned}$$

the Time Of Flight(TOF) is

$$\begin{aligned}\text{TOF} &= g + \sqrt{\frac{a^3}{\mu}} (\Delta E - \sin(\Delta E)) \\ &= 19.151688789 \text{ s} + \sqrt{\frac{(6681.607753295 \text{ km})^3}{3.986004418 \text{ km}^3/\text{s}^2}} (3.116590735 - 0.024999313) \\ &= 2696.602850754 \text{ s}\end{aligned}$$

If you assume that $\Delta E \approx \pi (= 3.141592654 \text{ rad})$

$$\text{TOF} = 19.151688789 \text{ s} + \sqrt{\frac{(6681.607753295 \text{ km})^3}{3.986004418 \text{ km}^3/\text{s}^2}} (3.141592654 \text{ rad}) = 2736.859048 \text{ s}$$

4. Rader readings determine that an object is located at

$$\vec{r} = 1.8\hat{K} \text{ DU} \quad \text{and} \quad \vec{v} = (0.8\hat{i} + 0.6\hat{K}) \text{ DU/TU}$$

Determine $p, e, I, \mathcal{Q}, \omega, \nu, u, \lambda_{true}$.

$$r = |\vec{r}| = 1.80 \text{ DU}$$

$$v = |\vec{v}| = 1.00 \text{ DU/TU}$$

$$\vec{h} = \vec{r} \times \vec{v} = 1.44 \text{ J} [\text{DU}^2/\text{TU}]$$

$$h = |\vec{h}| = 1.44 \text{ DU}^2/\text{TU}$$

$$\xi = \frac{v^2}{2} - \frac{\mu}{r} = \frac{1 \text{ DU}^2/\text{TU}^2}{2} - \frac{1}{1.8 \text{ DU}} = -0.06 \text{ DU}^2/\text{TU}^2$$

Find orbit element.

The eccentricity, e

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$$\begin{aligned}\vec{e} &= \frac{\left(v^2 - \frac{\mu}{r}\right)\vec{r} - (\vec{r} \cdot \vec{v})\vec{v}}{\mu} = \left(1.00^2 - \frac{1}{1.80}\right) \cdot (1.80\hat{R}) - \{(1.80\hat{R}) \cdot (0.8\hat{I} + 0.6\hat{K})\} \cdot ((0.8\hat{I} + 0.6\hat{K})) \\ &= -0.864\hat{I} + 0.152\hat{K} \\ e &= |\vec{e}| = 0.877\end{aligned}$$

The semi-major axis, a & semi-parameter, p

$$\begin{aligned}a &= -\frac{\mu}{2\xi} = -\frac{1DU^3/TU^2}{2 \cdot (-0.06DU^2/TU^2)} = 8.33 DU \\ p &= a(1 - e^2) = 8.33 DU \cdot (1 - 0.877^2) = 1.9232DU \\ \text{or } p &= h^2 = 2.0736DU\end{aligned}$$

The inclination, i

$$i = \arccos\left(\frac{h_k}{|\vec{h}|}\right) = 90^\circ$$

because h_k is 0.

The ascending node

$$\begin{aligned}\vec{n} &= \hat{R} \times \vec{h} = -1.44\hat{I} \\ \Omega &= \arccos\left(\frac{\hat{I} \cdot \vec{n}}{|\hat{I}||\vec{n}|}\right) = 180^\circ\end{aligned}$$

If $n_j < 0$ then $\Omega = 360^\circ - \Omega$

The argument of perigee, ω

$$\omega = \arccos\left(\frac{\vec{n} \cdot \vec{e}}{|\vec{n}||\vec{e}|}\right) = \arccos\left(\frac{(-1.44\hat{I}) \cdot (-0.862\hat{I} + 0.152\hat{K})}{0.144 \cdot 0.877}\right) = 9.98^\circ$$

The true anomaly, v

$$v = \arccos\left(\frac{\vec{e} \cdot \vec{r}}{|\vec{e}||\vec{r}|}\right) = \arccos\left(\frac{(-0.862\hat{I} + 0.152\hat{K}) \cdot 1.8\hat{K}}{0.877 \cdot 1.8}\right) = 80.0192^\circ$$

If $\vec{r} \cdot \vec{v} < 0$, then $v = 360^\circ - v$

$$\vec{r} \cdot \vec{v} = (1.8\hat{K}) \cdot (0.8\hat{I} + 0.6\hat{K}) = 1.08DU^2/TU$$

Because the dot product is positive, v is 139.67°

The longitude of perigee, $\tilde{\omega}_{\text{true}}$

$$\tilde{\omega}_{\text{true}} = \arccos\left(\frac{\hat{I} \cdot \vec{e}}{|\hat{I}||\vec{e}|}\right) = \arccos\left(\frac{\hat{I} \cdot (-0.862\hat{I} + 0.152\hat{K})}{0.877}\right) = 170.02^\circ$$

If $e_j < 0$, then $\tilde{\omega}_{\text{true}} = 360^\circ - \tilde{\omega}_{\text{true}}$

The argument of latitude, u

$$u = \arccos\left(\frac{\vec{n} \cdot \vec{r}}{|\vec{n}||\vec{r}|}\right) = \arccos\left(\frac{(-1.44\hat{I}) \cdot (1.8\hat{K})}{1.44 \cdot 1.8}\right) = 90^\circ$$

if $r_k < 0$ then $u = 360^\circ - u$

The true longitude, λ_{true}

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$$\lambda_{\text{true}} = \text{acos}\left(\frac{\hat{\mathbf{I}} \cdot \vec{\mathbf{r}}}{|\hat{\mathbf{I}}||\vec{\mathbf{r}}|}\right) = \text{acos}\left(\frac{\hat{\mathbf{I}} \cdot (1.8\hat{\mathbf{K}})}{1.8}\right) = 90^\circ$$

If $r_j < 0$ then $\lambda_{\text{true}} = 360^\circ - \lambda_{\text{true}}$

5. For the following orbital elements:

$$p = 0.42\text{DU}, e = 0.60, I = 90^\circ, u = 90^\circ, \Omega = 180^\circ, \omega = 260^\circ, v = 190^\circ, \lambda_{\text{true}} = 270^\circ$$

a. Express $\vec{\mathbf{r}}$ and $\vec{\mathbf{v}}$ for the satellite in the perifocal coordinate system along the unit vectors $\hat{\mathbf{P}}, \hat{\mathbf{Q}}, \hat{\mathbf{W}}$.

$$\vec{\mathbf{r}}_{\text{PQW}} = \begin{bmatrix} \frac{p \cos(v)}{1 + e \cos(v)} \\ \frac{p \sin(v)}{1 + e \cos(v)} \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{0.42\text{DU} \cdot \cos(190^\circ)}{1 + 0.82 \cdot \cos(190^\circ)} \\ \frac{0.42\text{DU} \cdot \sin(190^\circ)}{1 + 0.82 \cdot \cos(190^\circ)} \\ 0 \end{bmatrix} = \begin{bmatrix} -1.01\text{DU} \\ -0.18\text{DU} \\ 0 \end{bmatrix}$$

$$\vec{\mathbf{v}}_{\text{PQW}} = \begin{bmatrix} -\sqrt{\frac{\mu}{p}} \sin(v) \\ \sqrt{\frac{\mu}{p}} (e + \cos(v)) \\ 0 \end{bmatrix} = \begin{bmatrix} -\sqrt{\frac{1\text{DU}^3\text{TU}^2}{0.42\text{DU}}} \sin(190^\circ) \\ \sqrt{\frac{1\text{DU}^3\text{TU}^2}{0.42\text{DU}}} (0.82 + \cos(190^\circ)) \\ 0 \end{bmatrix} = \begin{bmatrix} 0.27\text{DU/TU} \\ -0.59\text{DU/TU} \\ 0 \end{bmatrix}$$

b. By a suitable coordinate transformation technique, express $\vec{\mathbf{r}}$ and $\vec{\mathbf{v}}$ in the geocentric equatorial system in the $\hat{\mathbf{I}}, \hat{\mathbf{J}}, \hat{\mathbf{K}}$ axis.

Rotation matrix $\mathbf{R} = \text{ROT3}(-\Omega)\text{ROT1}(-i)\text{ROT3}(-\omega)$

$$\mathbf{R} = \begin{bmatrix} \cos(-\Omega) & \sin(-\Omega) & 0 \\ -\sin(-\Omega) & \cos(-\Omega) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(-i) & \sin(-i) \\ 0 & -\sin(-i) & \cos(-i) \end{bmatrix} \begin{bmatrix} \cos(-\omega) & \sin(-\omega) & 0 \\ -\sin(-\omega) & \cos(-\omega) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \cos(\Omega) \cos(\omega) - \sin(\omega) \cos(i) \sin(\omega) & -\cos(\Omega) \sin(\omega) - \sin(\Omega) \cos(i) \cos(\omega) & \sin(\Omega) \sin(i) \\ \sin(\Omega) \cos(\omega) + \cos(\Omega) \cos(i) \sin(\omega) & -\sin(\Omega) \sin(\omega) + \cos(\Omega) \cos(i) \cos(\omega) & -\cos(\Omega) \sin(i) \\ \sin(i) \sin(\omega) & \sin(i) \cos(\omega) & \cos(i) \end{bmatrix}$$

$$\mathbf{R} = \begin{bmatrix} \text{IJK} \\ \text{PQW} \end{bmatrix} = \begin{bmatrix} 0.17 & -0.98 & 0 \\ 0 & 0 & 1 \\ -0.98 & -0.17 & 0 \end{bmatrix}$$

$$\vec{\mathbf{r}}_{\text{IJK}} = \begin{bmatrix} \text{IJK} \\ \text{PQW} \end{bmatrix} \vec{\mathbf{r}}_{\text{PQW}} = \begin{bmatrix} 0.17 & -0.98 & 0 \\ 0 & 0 & 1 \\ -0.98 & -0.178 & 0 \end{bmatrix} \begin{bmatrix} -1.01\text{DU} \\ -0.18\text{DU} \\ 0 \end{bmatrix} = \begin{bmatrix} 0.0047\text{ DU} \\ 0 \\ 1.0204\text{ DU} \end{bmatrix}$$

$$\vec{\mathbf{v}}_{\text{IJK}} = \begin{bmatrix} \text{IJK} \\ \text{PQW} \end{bmatrix} \vec{\mathbf{v}}_{\text{PQW}} = \begin{bmatrix} 0.17 & -0.98 & 0 \\ 0 & 0 & 1 \\ -0.98 & -0.178 & 0 \end{bmatrix} \begin{bmatrix} 0.27\text{DU/TU} \\ -0.59\text{DU/TU} \\ 0 \end{bmatrix} = \begin{bmatrix} 0.6241\text{ DU/TU} \\ 0 \\ -0.1643\text{ DU/TU} \end{bmatrix}$$